

TL;DR

IMPLEMENT lets a **frozen LLM** plan imaginatively in visual embodied environments: a lightweight world model *perceives visually but predicts symbolically*, Monte-Carlo rollouts capture uncertainty, and Meta In-Context Learning adapts it online. **The frozen LLM and the learned world model co-evolve during test time to achieve “in-context policy iteration”**.

Motivation

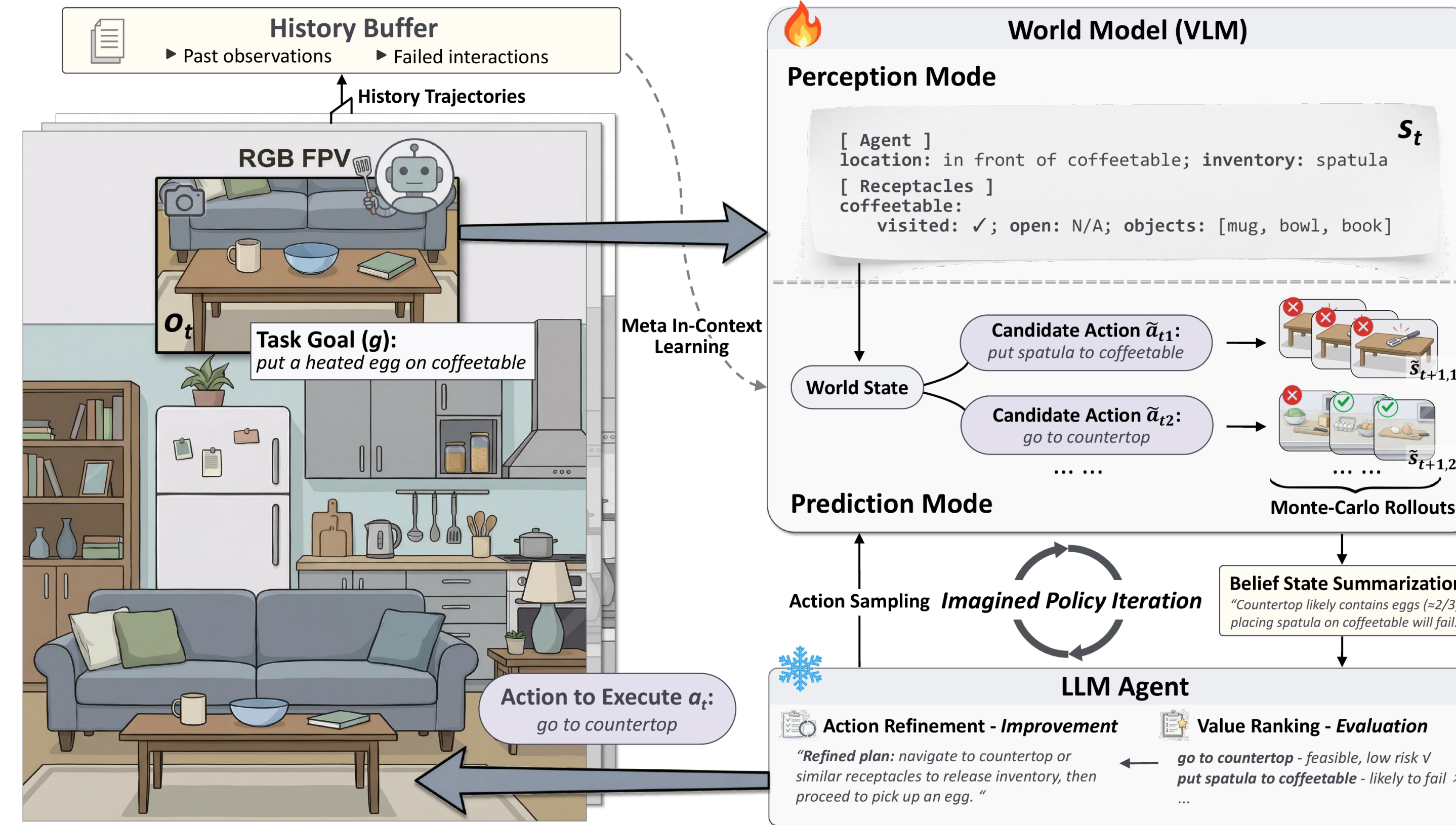
Planning needs a **predictive model of dynamics**. LLMs learn symbolic rules from pretraining, but **embodied worlds are different**:

- Sparse visual evidence** — pixels are high-dimensional, semantics are sparse.
- Partial observability** — uncertainty compounds over long horizons.
- Distribution shift** — new layouts, objects and affordances every episode.

Gap in prior work. **Finetuning** is costly and locks out frontier models; **test-time scaling** has no dynamics model; **LLM-as-world-model** leans on language priors and keeps policy and model static.

Key insight — **perceive visually, predict symbolically**. Pair a frozen LLM with a learned world model and let them co-evolve in context.

Framework Overview



IMPLEMENT: a **frozen LLM** plans over futures imagined by a **learned world model**; the history buffer drives Meta In-Context Learning (MetaICL).

One environment step: imagine → evaluate → refine

1 Perceive

[PER] mode turns the RGB view into an object-centric symbolic state: location, inventory, receptacle & object attributes. Updated every step. Maintain only confirmed facts.

2 Imagine

The LLM proposes candidate actions and a **help** flag. **[PRE]** mode rolls out $M = 10$ futures per candidate with temperature sampling, conditioned on history τ (MetaICL).

3 Evaluate

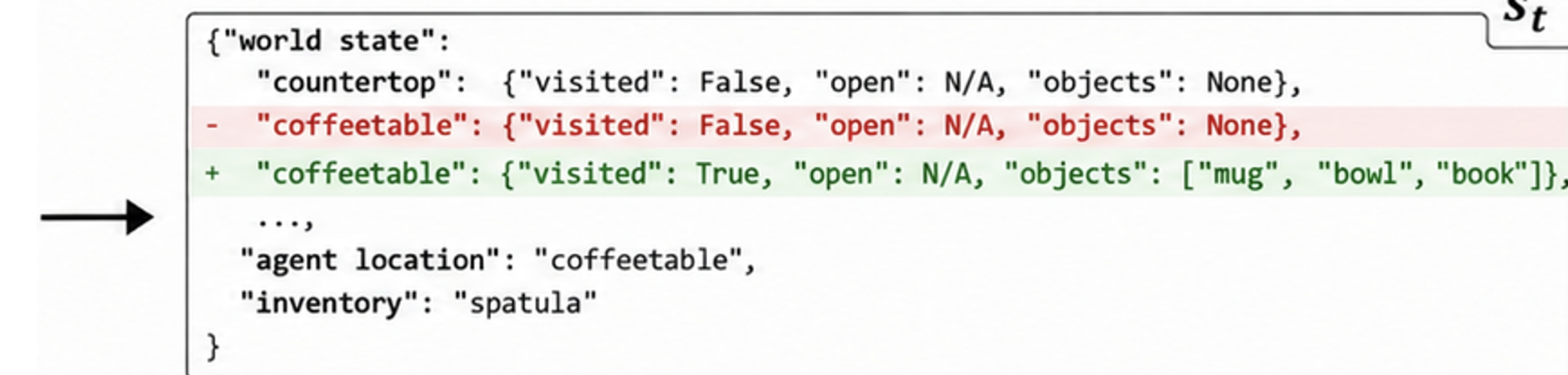
Futures are compressed into a textual **belief state** (“countertop likely has eggs $\approx 2/3$ ”); the LLM ranks candidates by imagined outcomes — an implicit value estimation.

4 Refine or commit

Refine for another round, or commit the action to the environment (avg. 2.4 rounds). Executed outcomes enter the history buffer and sharpen the world model in context.

Method

Symbolic world model — one VLM, two modes



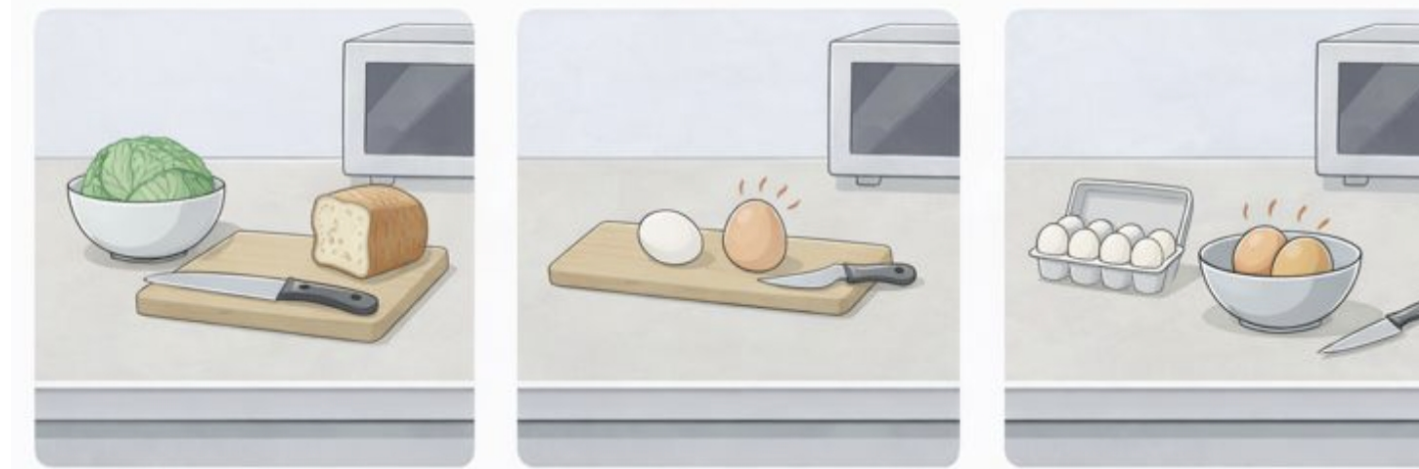
s_t — symbolic state after “go to coffeetable”: only the visited receptacle and location change.

Object-centric state: location · inventory · receptacle/object attributes (visited, open, heated, containment...).

Mode tokens: **[PER]** updates the state from the image; **[PRE]** predicts only action-contingent changes.

Self-labeled training: Qwen2.5-VL-7B + LoRA on $\approx 80K$ transitions from mostly random exploration — no rewards, no high-quality expert demos.

Imaginative planning (IP) + Meta In-Context Learning (MetaICL)



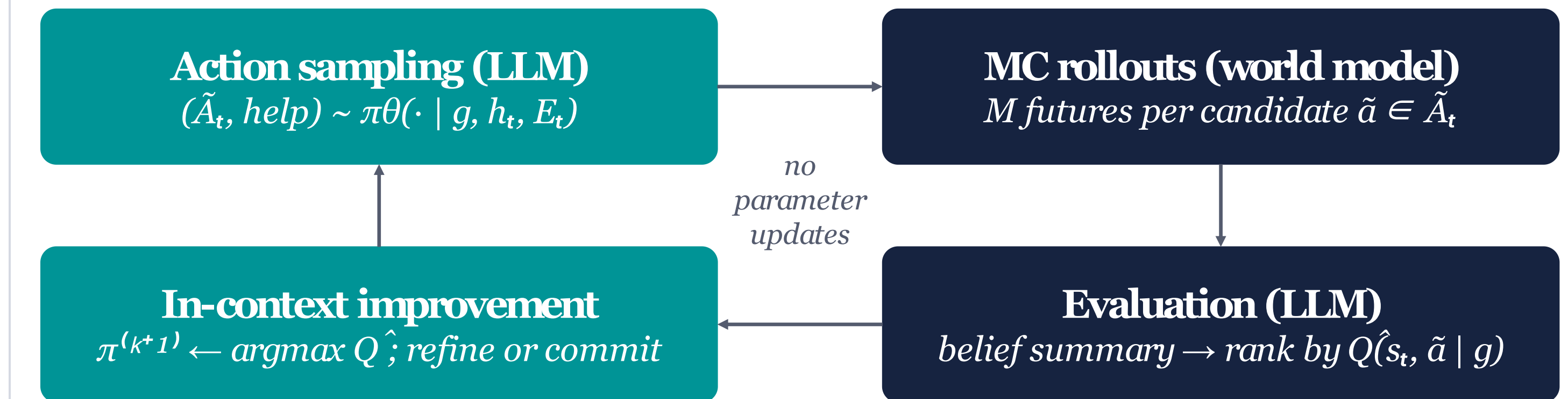
Monte-Carlo state prediction: sample M futures per candidate action with non-zero temperature instead of one point estimate.

Belief-state summarization: futures → short text with outcome frequencies, keeping the planner context compact and intuitive.

MetaICL: condition on trajectory context τ (past observations + failed actions) → the world model **self-evolves online** without parameter updates.

With history: failed-action prediction **63.0 → 95.3 %**, unobserved-object recall **56.0 → 70.6 %**.

Imagine → evaluate → refine = policy iteration at test time

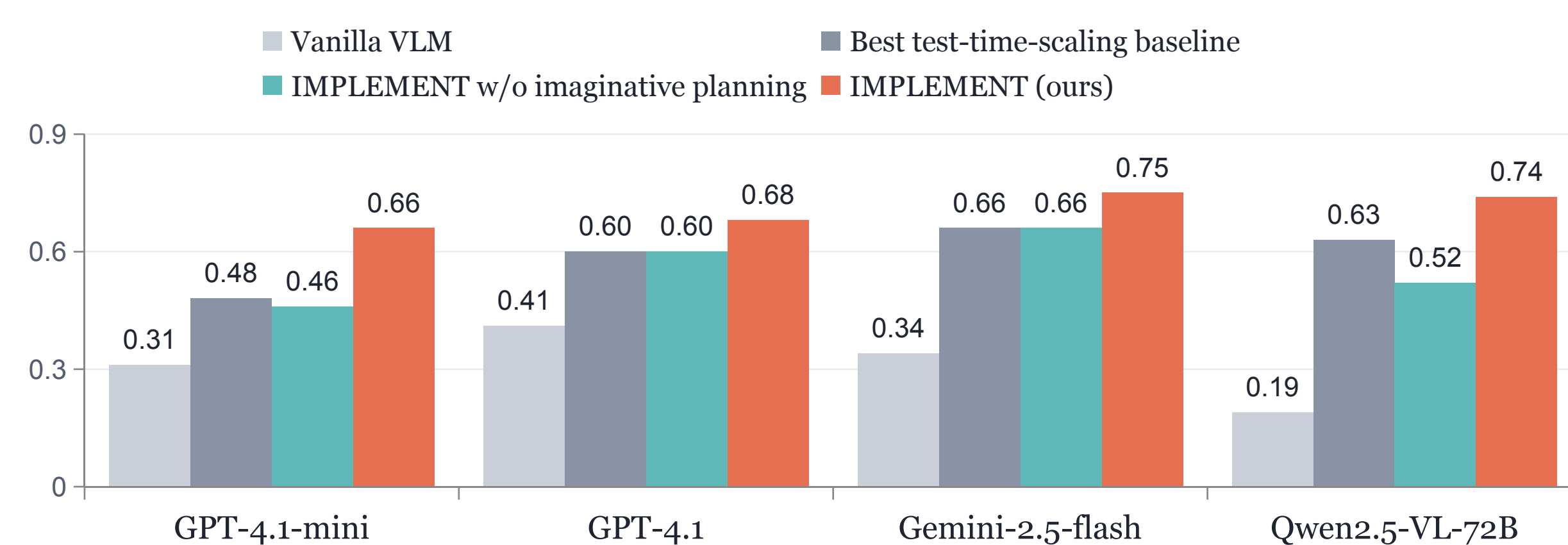


Generalized policy iteration in context: rollouts + LLM reasoning form an implicit value landscape — no parametric value function.

help flag = adaptive termination: only 2.4 rounds per step on average (cap 5).

Main Results

SR@1 on 134 out-of-distribution ALFWorld tasks



+18 pp

SR@1 vs. best frozen baseline (GPT-4.1-mini)

0.89

SR@12, Qwen2.5-VL-72B; finetuned EMMA: 0.82

+23.1 pp

SR@1 vs. SimuRA, same perception (Qwen3.5-Plus)

IMPLEMENT vs. strongest frozen baseline

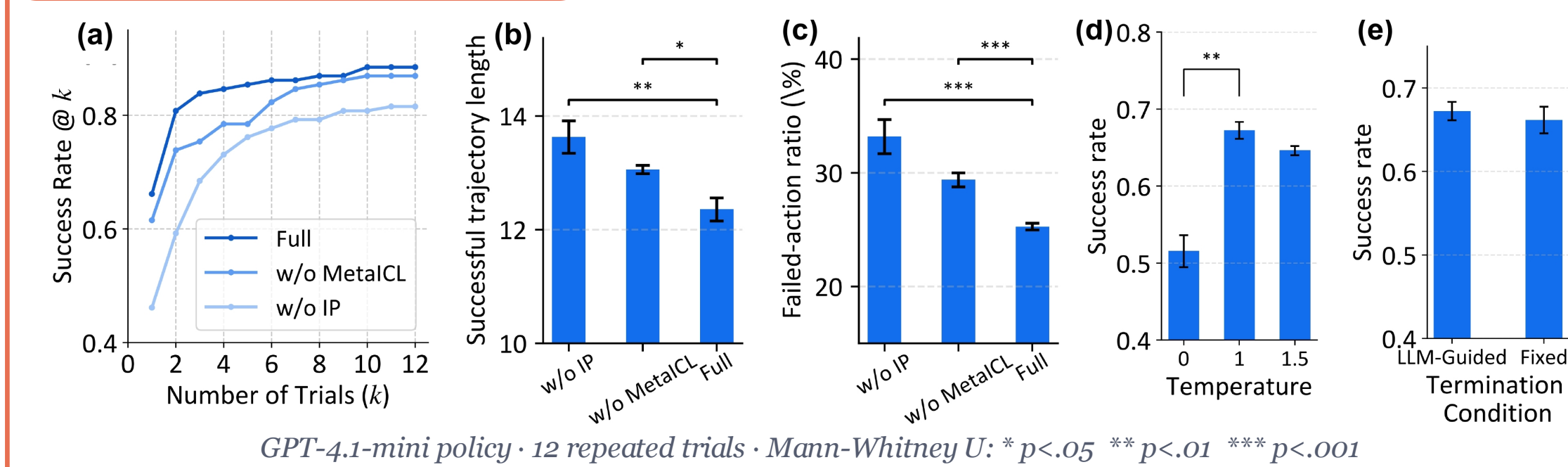
Policy LLM (frozen)	SR @ 1		SR @ 6		SR @ 12	
	Base [†]	Ours	Base [†]	Ours	Base [†]	Ours
GPT-4.1-mini	.48	.66	.74	.86	.81	.88
GPT-4.1	.60	.68	.81	.82	.82	.85
Gemini-2.5-flash	.66	.75	.78	.86	.78	.88
Qwen2.5-VL-72B	.63	.74	.76	.85	.79	.89
EMMA (finetuned)*	.19	—	.58	—	.82	—

[†] Base = best of ReAct / Reflexion / Self-consistency / SimuRA for the same policy LLM and trial budget (may differ per cell). * EMMA is finetuned on privileged demonstrations; shown for reference only.

Wins on every backbone and every budget; gains are largest at low budgets → superior sample efficiency.

Same success, less compute: 3 trials × 23.6 s vs. 12 trials × 7.5 s without imagination; only +8.8 % tokens.

Ablations



(a) IP and MetaICL both raise success at every budget; the gap is largest for $k \leq 5$.

(b, c) Without IP: longer trajectories and far more failed actions ($p < 0.001$).

(d) Temperature 0 drops sharply — Monte-Carlo rollouts are essential.

(e) LLM-guided termination matches fixed 5 rounds with only 2.4 on average.

Takeaways

- Ground reasoning in a world model.** Perceive visually, predict symbolically: a 7B world model gives any frozen LLM dynamics- and uncertainty-aware planning.
- Co-evolve policy and model at test time.** Imagine → evaluate → refine is policy iteration in context; MetaICL learns from every failure.
- Consistent gains where it matters.** Wins on 4 backbones × 3 budgets, beats finetuned SOTA, strongest in the low-trial regime.